

Subject Description Form

Subject Code	AAE2005
Subject Title	Electrics and Electronic for Aeronautical Engineering
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject will provide students with 1. The fundamental knowledge of electrics and electronics in aeronautical engineering; 2. Basic electrical and electronic devices on aircraft; and 3. Design and development of basic electronic device.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand the fundamental concepts of electrical circuits in aeronautical engineering; b. Understand the principle of basic electric devices in aeronautical engineering; c. Apply the appropriate techniques to solve problems in electrical circuits; and d. Design basic electronic devices with the basic knowledge.
Subject Synopsis/ Indicative Syllabus	Basic law: Circuit elements; Kirchhoff's Current Law (KCL); Kirchhoff's Voltage Law (KVL); Ohm's Law; series and parallel circuits Semiconductor: Intrinsic semiconductor, p-type semiconductor, n-type semiconductor. Diode and circuits : Formation of PN junction, the working principle of diode and its basic circuits. Transistor and circuits: the current amplification effect of BJT; carrier motion analysis in BJT. Logical operation and circuits: basic logic operation and devices, Combinational logic operation and circuits, Sequential logic operation and circuits Other basic electronic circuits: Filter, ADC, Decoder, Counter and etc.

Teaching/Learning Methodology	The key concepts and techniques covered in this subject are discussed in lectures and tutorials. The lectures emphasise on fundamental understanding and practical problem-solving techniques. To strengthen understanding of fundamental knowledge of electric and electronic in the field of aeronautical engineering, students have the chances to conduct hands-on exercises in both lectures and tutorials. Furthermore, individual assignments or tests consisting of essays and the numerical problems are involved to allow students recognise their level of understanding and create evidence of learning.																																													
	Teaching/Learning Methodology		Intended subject learning outcomes to be covered																																											
			a	b	c	d																																								
	1. Lecture		✓	✓	✓																																									
	2. Tutorial		✓	✓	✓	✓																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">Weighting %</td><td colspan="4">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>1. Assignment</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Lab</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Case study presentation</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Examination</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table>						Specific assessment methods/tasks	Weighting %	Intended subject learning outcomes to be assessed				a	b	c	d	1. Assignment	20%	✓	✓	✓	✓	2. Lab	20%	✓	✓	✓	✓	3. Case study presentation	20%	✓	✓	✓	✓	4. Examination	40%	✓	✓	✓	✓	Total	100%				
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2. Lab	20%	✓	✓	✓	✓																																									
3. Case study presentation	20%	✓	✓	✓	✓																																									
4. Examination	40%	✓	✓	✓	✓																																									
Total	100%																																													
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:																																													
	Overall assessment:																																													
	0.6 x continuous assessment + 0.4 x examination																																													
	Assignment, lab and case study presentation are used to make up the continuous assessment and the final examination is included to evaluate and assess the students' learning outcomes in this course. The continuous assessment and the final exam are conducted at different times in the semester to consolidate students' knowledge in lectures and tutorials. They are appropriate in assessing intended learning outcomes.																																													

Student Study Effort Expected	Class contact:	
	▪ Lecture/Tutorial	39 Hrs.
	Other student study effort:	
	▪ Literature Review and Self-learning	46 Hrs.
	▪ Assignments	26 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. Giorgio Rizzoni, “Fundamentals of Electrical Engineering”, New York: McGraw- Hill, 2009. 2. Giorgio Rizzoni, James Kearns, “Principles and Applications of Electrical Engineering” 6th Edition, Boston: McGraw-Hill Higher Education, 2018. 3. “E Electrics and Electronics. ATPL Ground Training Series”, CAE Oxford Aviation Academy.	

Jun 2024